IQP Generative Models for Strong Gravitational Lensing Images

Anonymous Author(s)
Affiliation
Address
email

Abstract

Sampling from instantaneous-quantum-polynomial (IQP) circuits is believed to be classically hard [3, 4], while the expectation values a maximum-mean-discrepancy (MMD) loss needs stay classically computable, so an IQP Born machine can be trained classically and deployed on quantum hardware [12]. A recent correlation–complexity map predicts which datasets suit such models, and on it DeepLense strong-lensing images are the most IQP-compatible dataset we measured ($I_{\mathrm{QCLI}} = 0.47$, $I_{\mathrm{CCI}} = 0.87$), above the turbulence data of the paper that introduced it. Given *learned* latent directions, $\theta = \theta^0 + \Phi^\top z$, a 10-qubit IQP Born machine trained on lens crops beats an RBM and a DCGAN on the paper's PDF–JS and flux-aligned Conv2D-MMD^2, at $N = 11$ and $N = 100$ training images with 2–39$\times$ fewer parameters, and reconstructs unseen images at correlation 0.97. Three caveats bound it. The reference paper's own fixed latent protocol yields only noise on images, a structure-matched permutation null still scores $I_{\mathrm{QCLI}} = 0.42$, and the DCGAN still leads on raw Conv2D-MMD^2 at $N = 100$. Everything is exact classical simulation at a single seed and $n = 10$ qubits, so we claim no quantum advantage.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard transform of a phase function. Exact classical sampling from them would collapse the polynomial hierarchy [3], and the hardness is conjectured to survive approximate sampling in the average case [4, 5], though sufficient noise restores classical simulability [11]. The same commuting structure keeps the expectation values in an MMD loss classically computable, so such a model can be *trained classically and deployed on quantum hardware* [12, 13].

Whether real scientific data sits where that is useful is an empirical question, and a recent proposal [9] approaches it with two ingredients. (i) A *correlation–complexity map* placing a dataset on two coordinates, the quantum correlation-likeness indicator I_{QCLI}, how much of its Walsh order spectrum sits at high order as an IQP output would, and the classical correlation-complexity indicator I_{CCI}, how much of its total correlation the best pairwise tree misses [16, 14]; the high-I_{QCLI}, high-I_{CCI} corner is argued to be where IQP circuits do well. (ii) A latent-adaptation protocol (their Algorithm 1) that fits a shared core to one anchor sample, then adapts only a $d_{\mathrm{lat}} = 50$ latent block of Hamming-ordered Pauli strings per sample. Both were demonstrated on one dataset, 2D turbulence snapshots [8] at $n = 18$ qubits.

We take both to simulated dark-matter strong-lensing images from DeepLense [1, 2] and report three results. (1) On the reference paper's own map, lens crops land at the most IQP-compatible position of any dataset we measured, above the turbulence data the map was built to recommend. (2) A 10-qubit IQP Born machine trained on those crops beats the paper's own RBM and DCGAN baselines on

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PDF–JS and flux-aligned Conv2D-MMD^2 at both training-set sizes with far fewer parameters. (3) It does so only after one change to the reference protocol, whose fixed Walsh latent block gives noise on images until its directions are made *learnable*. Appendices B–D take the same tests back to the paper's own data, where its turbulence result does not reproduce under our pre-registered convention and two internal claims come out mixed.

2 Setup

Data and encoding. We use the DeepLense `Model II` simulation set (80,194 train / 8,910 validation images, 64×64, three substructure classes pooled) [1]. Each image is centroid-aligned by an integer shift (offset 3.4 ± 1.6 px) and cropped to the central 32×32 window, keeping 99.4% of the light, and the crop becomes a probability mass function (pmf) $p(k_x, k_y)$ over $1{,}024$ cells on $n = 10$ qubits (App. A). All 2^{10} generators Z_s are used, so $\theta \in \mathbb{R}^{1024}$ and q_θ follows from two length-1024 fast Walsh–Hadamard transforms. Generated pmfs return to 64×64 with a fixed flux $S = 124.92$ and an offset from the empirical distribution.

Latent parameterisations. The reference protocol keeps a fixed $d_{\mathrm{lat}} = 50$ block of θ as the per-image latent and freezes the other 974 coordinates, fitted to one anchor image (Algorithm 1, sequential). We compare it against (i) the same fixed block trained *jointly* across images and (ii) our *learned-direction* model

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M \times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

with $M \in \{2, 4, 8\}$, each row of Φ L^2-normalised to $\sqrt{1024}$, and $\theta^0, \Phi, \{z_i\}$ trained jointly under an L^2 penalty on z of 10^{-3}, raised to 0.3 for the $M{=}8$ arm. The three objectives are the paper's multi-bandwidth Gaussian MMD kernel, a flat-Walsh-weight MMD estimable from hardware shots, and exact KL. For unconditional generation z is drawn from a fitted Gaussian or resampled from the trained z_i with jitter.

Baselines and metrics. We compare only against the two baselines the reference paper uses, a restricted Boltzmann machine (RBM; 10 visible and 1,000 hidden units, persistent contrastive divergence with one Gibbs step, PCD-1 [7, 15]; 11,010 parameters) and a deep convolutional generative adversarial network (DCGAN) [10] on 32×32 crops (199,837 parameters), trained on the same data with the same seed. Metrics are the paper's two, the probability-density Jensen–Shannon divergence between intensity histograms (PDF–JS) and Conv2D-MMD^2, a biased MMD estimate in the feature space of a fixed random three-layer convolutional encoder under a median-heuristic Gaussian kernel [6]. Every IQP arm and the RBM share the flux scalar S while the DCGAN learns its own scale, so raw Conv2D-MMD^2 is partly a brightness comparison and we also report a flux-aligned variant, flagged as post-hoc. Every arm generates 200 images scored against the same 200 held-out images at the reference training-set sizes $N \in \{11, 100\}$, with a single seed (seed 0) throughout.

3 Results

3.1 Lens images are the most IQP-compatible dataset on the map

Figure 1(a) recomputes the map on our data. Encoded as the reference paper encodes turbulence (18-bit (x, y, z) tuples, $K = 4{,}096$ sampled tuples per image), lens images score $I_{\mathrm{QCLI}} = 0.4715$ and $I_{\mathrm{CCI}} = 0.8677$, higher in I_{QCLI} than the paper's own turbulence data at the same K (0.023), binarised MNIST (0.014) or BinaryBlobs (0.032, both at $K = 10^4$), and squarely inside the corner the map calls IQP-compatible, a stronger recommendation than the dataset the map was introduced to justify.

Two caveats temper that reading. A permutation null that keeps each image's intensity histogram exactly and destroys all spatial structure still scores $I_{\mathrm{QCLI}} = 0.4154$ and *raises* I_{CCI} to 0.8782, leaving only $+0.056$ in I_{QCLI} from lens structure, and the paper's turbulence data behaves the same way at the same K (0.023 against a null of 0.011), so at this encoding the map largely tracks the intensity histogram. The 10-bit crop pmf we train on does separate from both nulls in both coordinates (0.473 against 0.178, 0.431 against 0.963), and we therefore read a favourable map position as necessary rather than sufficient, settling the question by training.

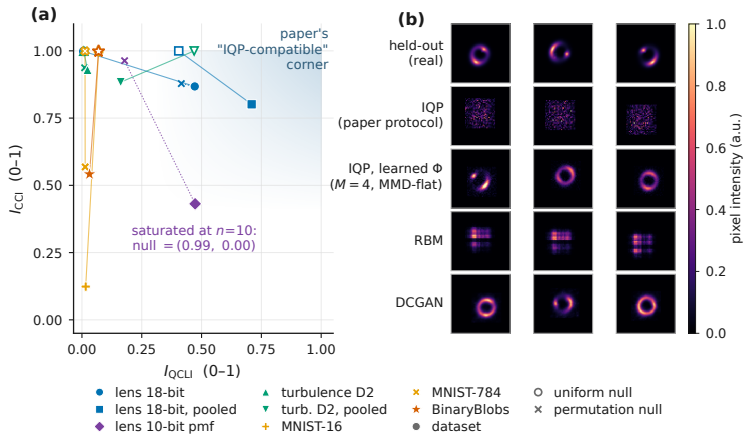


Figure 1: **(a)** Correlation–complexity map. Datasets (filled), i.i.d.-uniform nulls at the same (n, K) (open), permutation nulls (crosses; ours, not in the reference paper). The shaded corner is the reference paper's *qualitative* IQP-compatible region. **(b)** First three unconditional samples per model, $N{=}100$, no selection, shared 0–1 colour scale (at most 13 pixels per row are clipped).

3.2 Why the reference protocol needs learned directions

Run with the paper's sequential protocol, using our low-variance latent block, Algorithm 1 produces noise (Fig. 1(b), second column; the Hamming-ordered variant is in App. E). Representability is not the cause, since the full 1,024-phase circuit fits a *single* aligned crop to correlation 0.9999 (KL), 0.99997 (flat MMD) and 0.9962 (the paper's kernel). The protocol is. With the core frozen on one anchor, the 50 fixed latent coordinates reconstruct every other training image at correlation 0.47, because fixed Walsh directions act as an XOR-shift interference pattern that cross-fades pixels instead of moving lens structure, which suits a drifting time series and not an i.i.d. image set.

3.3 Learned latent directions generalise

Equation (1) repairs this. To test generalisation we freeze θ^0 and Φ and optimise only z on 200 *held-out* images. At $N{=}100$ the best arm reaches mean correlation 0.968 (minimum 0.851) against a nearest-training-image reference of 0.970. The gain comes from learning the directions, not the extra parameters, since in a separate controlled sweep at $N{=}20$ (Fig. 2(a)) the trained model reaches 0.911 while a random Φ of the same size gives 0.672 and no latent gives 0.671. The flat-MMD kernel, which is estimable from hardware shots, generalises best (0.968 at $M{=}4$) and the paper's own multi-bandwidth kernel worst (0.74–0.79), with KL in between (0.914 at $M{=}2$). Useful M stays small, as that sweep peaks at $M{=}2$–4 and collapses to 0.10 by $M{=}8$ once the required $\|z\|$ exceeds the model's coherence radius ($\approx$ 0.03–0.05). At $M{=}2$ all prior draws are rings of rms radius 9.40 ± 0.21 px against 9.21 ± 0.15 for real data, though under-dispersed in ellipticity (0.048 vs. 0.099).

3.4 Paper-style comparison and data scaling

In Table 1 the best IQP arm wins PDF–JS at both training-set sizes (0.00095 and 0.00092) and wins the flux-aligned Conv2D-MMD^2 at both (0.0279 and 0.0275), with $2\times$ fewer parameters than the RBM and $39\times$ fewer than the DCGAN. Two of five IQP arms fall behind the DCGAN on PDF–JS at $N{=}11$ and three of five fall behind the RBM at $N{=}100$, and on raw Conv2D-MMD^2 the IQP arms *lose* at both sizes, to the RBM at $N{=}11$ (0.0741 against 0.1134) and to the DCGAN at $N{=}100$ (0.0371 against 0.0861); that ordering is partly an artefact of the imposed flux scalar, since it reverses once flux is aligned. The paper's data-scaling claim holds only in part. Our single-seed DCGAN is *better* at $N{=}11$ than at $N{=}100$ on PDF–JS, the opposite of the low-data GAN collapse the paper reports, and in Fig. 2(c) it leads already at $N{=}10$ and swings $3\times$ non-monotonically while the IQP arm stays flat for $N \geq 30$ (App. E).

Table 1: Paper-style comparison on lens crops, IQP against the two baselines the reference paper uses. Lower is better and **bold** marks the best entry in a column. Learned-direction parameter counts are $N{=}11$ / $N{=}100$, and $\mathrm{MMD}^2_{\mathrm{al}}$ is the flux-aligned variant (post-hoc). The last row is an empirical floor from scoring one held-out half against the other, not a model. Single seed.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	$\mathrm{MMD}^2_{\mathrm{al}}$	PDF–JS	MMD^2	$\mathrm{MMD}^2_{\mathrm{al}}$
IQP, paper protocol (Alg. 1)	1,024	0.00609	0.2032	0.1837	0.00697	0.2140	0.2233
IQP, fixed basis, joint	1,024	0.00133	0.1416	0.1141	0.00099	0.1041	0.0617
IQP, learned dir. $M{=}2$ (KL)	3,094 / 3,272	0.00175	0.1673	0.0559	0.00302	0.1356	0.0866
IQP, learned dir. $M{=}4$ (flat MMD)	5,164 / 5,520	0.00196	0.1134	**0.0279**	0.00239	0.0861	**0.0275**
IQP, learned dir. $M{=}4$ (paper kernel)	5,164 / 5,520	**0.00095**	0.1168	0.0679	**0.00092**	0.1189	0.0876
RBM [15]	11,010	0.00632	**0.0741**	0.1240	0.00195	0.0752	0.1144
DCGAN [10]	199,837	0.00183	0.2165	0.1241	0.00763	**0.0371**	0.0694
Held-out split-half (floor)	n/a	*0.00003*	*0.0040*	*0.0069*	*0.00003*	*0.0040*	*0.0069*

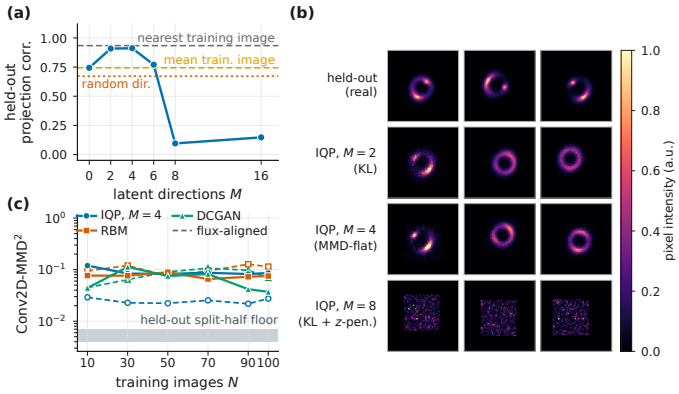


Figure 2: Panels (a) and (b) are different experiments. **(a)** Controlled latent-width sweep ($N{=}20$ mechanism test), held-out projection correlation against the number of learned directions M, with that sweep's own reference levels (nearest training image 0.934, mean training image 0.744, random directions 0.672); the $M{=}8$ and 16 points are the best over a learning-rate sweep, i.e. favourable to the model. **(b)** Round-3b arms at $N{=}100$, held-out crops (top), then $M{=}2$ (KL), $M{=}4$ (flat MMD), $M{=}8$ (KL + z-penalty). **(c)** Conv2D-MMD^2 vs. training-set size (solid raw, dashed flux-aligned).

4 Discussion and limitations

Findings. IQP sampling is classically hard while the MMD training signal is not, which is what makes classical training with quantum deployment plausible. The reference paper's map puts strong-lensing crops at the most IQP-compatible position we measured, and a 10-qubit IQP Born machine trained on those crops beats an RBM and a DCGAN on PDF–JS and on flux-aligned Conv2D-MMD^2 at both training-set sizes with far fewer parameters. Making the latent directions Φ trainable is the one change that unlocks it.

Limitations. Nothing here is evidence of quantum advantage, because every circuit is simulated exactly at $n = 10$ qubits, where the classical cost is two length-1024 transforms, and the hardness results describe the model class at scale rather than our instances. All numbers are single seed; the DCGAN's $3\times$ swing in Fig. 2(c) sets the scale of the effects we discuss. The flux S is imposed, which makes raw Conv2D-MMD^2 partly a brightness metric, and the aligned variant is post-hoc. The map's indicators barely separate real data from permutation nulls. We compare only against the paper's two baselines, and zero-learning and PCA controls we ran match or exceed the IQP arms on some physical-quantity metrics. Code and per-arm result tables will be released with the paper.

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A Pipeline

Figure 3 summarises the encoding, the latent parameterisation and the evaluation.

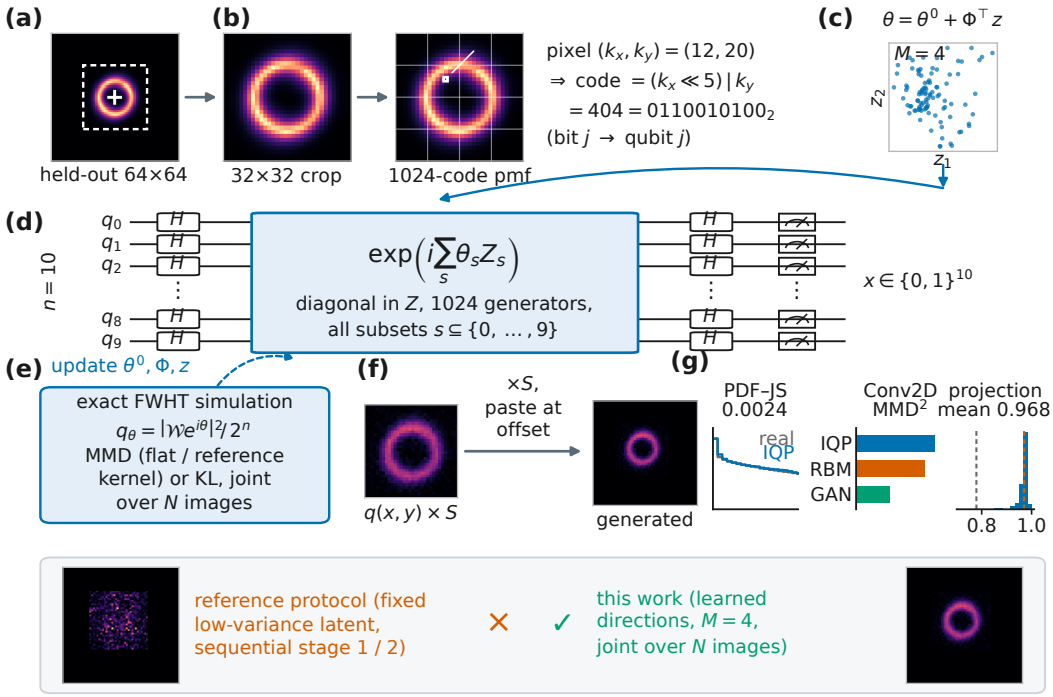


Figure 3: Pipeline. **(a)** A held-out 64×64 image, white cross at its intensity centroid and dashed box the 32×32 crop window. **(b)** The aligned crop read as a pmf over $1{,}024$ cells, where the example pixel $(k_x, k_y) = (12, 20)$ carries code $404 = 0110010100_2$ and bit j addresses qubit j. **(c)** The first two learned latent coordinates of the 100 training images at $M = 4$. **(d)** The IQP circuit, an H layer, one diagonal phase $\exp(i \sum_s \theta_s Z_s)$ over all $1{,}024$ subsets s, a second H layer and measurement. **(e)** Exact simulation through $q_\theta = |\mathcal{W} e^{i\theta}|^2 / 2^n$, trained jointly over the N images by MMD (flat or reference kernel) or by KL, the dashed arrow updating θ^0, Φ and z. **(f)** A trained $q(x, y)$ scaled by S and pasted at a sampled offset. **(g)** The intensity PDF of real against IQP samples (PDF–JS 0.0024), Conv2D-MMD^2 at N=100 (IQP 0.086, RBM 0.075, DCGAN 0.037), and projection correlations over 200 held-out images (mean 0.968). The bottom row contrasts a reference-protocol sample with an M=4 sample from this work.

B Replication of the reference paper's turbulence experiment

We attempted a direct replication on the reference paper's own dataset (2D cylinder-wake turbulence D2 [8], $n = 18$ qubits, 11 anchor snapshots). Under our pre-registered evaluation convention the result is **not reproduced**, since in the 11-snapshot regime the RBM and a 1.2M-parameter DCGAN beat our IQP on *both* of the paper's metrics. Two findings qualify this. Under a last-write-wins aggregation convention, never specified in the paper, our IQP's PDF–JS values do land close to the published ones (0.0322 and 0.0476 against 0.0238 and 0.0221), but the RBM still wins. Conv2D-MMD^2 is also saturated there, since ground truth shuffled in pixel space already scores 0.704, below every model in the paper's Table 2 (0.90–1.47), and the unseen-time synthesis task is solvable by linear interpolation (correlation 0.9998). Our anchor-fit ceiling (correlation 0.69 sampled, 0.77 exact) traces to a frozen θ_{lat} in stage 1 (≈ -0.08), a 5×10^4-sample pmf (≈ -0.04) and a 0.88–0.89 model-family ceiling, not to our implementation.

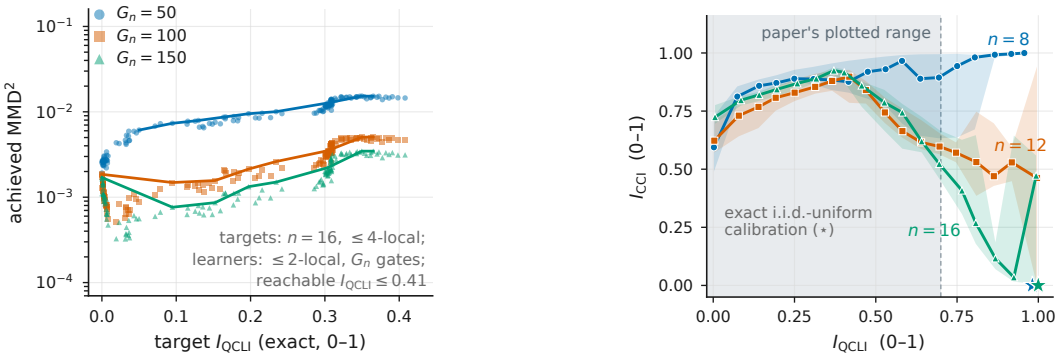


Figure 4: **(a)** Support-mismatch study at $n = 16$, with 140 target circuits ($\leq$ 4-local) spanning exact $I_{\mathrm{QCLI}} \in [0, 0.408]$ fitted by $\leq$ 2-local learners of 50, 100 and 150 gates (exact MMD, Adam, 5,000 steps). More gates lower the whole envelope (claim (b), reproduced), while within the reachable range the achieved MMD^2 *rises* with target I_{QCLI}, the opposite of claim (a). **(b)** Circuits optimised to maximise I_{QCLI} only, $n = 8, 12, 16$, 3 seeds and 2 gain settings per n, 3,333 steps each and about 20,000 logged points per n. I_{CCI} never enters the objective.

C Support mismatch (reference paper Fig. 3)

The reference paper's Fig. 3 claims (a) that the upper envelope of achievable MMD falls as the target's I_{QCLI} rises, because the support-induced floor shrinks, and (b) that the envelope drops overall as the learner gets more gates (Fig. 4(a)). Claim (b) reproduces, as the median achieved MMD^2 falls $9.6 \times 10^{-3} \to 2.2 \times 10^{-3} \to 1.45 \times 10^{-3}$ across the three learners. Claim (a) runs the other way inside the range we can reach, since the achieved MMD^2 *rises* as the target's exact I_{QCLI} goes from 0 to 0.41 (Spearman $\rho = +0.98/+0.90/+0.81$), a trend that also holds for a control trained on exact pmfs. The paper's premise still holds mechanistically ($\rho \leq -0.98$ for the Proposition II.2 floor proxy), but here the achieved MMD is dominated by trainability rather than by support mismatch. Our target family saturates at exact $I_{\mathrm{QCLI}} \approx 0.41$, and at 0–0.064 when estimated from 10^4 samples as the paper does, so the regime where claim (a) is asserted lies outside what we could construct.

D The I_{QCLI}–I_{CCI} ceiling

We also reproduced the experiment in which maximising I_{QCLI} alone raises I_{CCI} as a by-product (Fig. 4(b)). The coupling reproduces at low I_{QCLI} (Spearman $\rho = +0.83/+0.76/+0.95$ for $n = 8/12/16$ restricted to $I_{\mathrm{QCLI}} \leq 0.3$) but reverses above $I_{\mathrm{QCLI}} \approx 0.38$ for $n = 12, 16$ (global $\rho = -0.53, -0.57$). The reversal is forced by the definitions, because the exactly uniform distribution maximises I_{QCLI} (0.982/0.998/1.000 for $n = 8/12/16$) and has $I_{\mathrm{CCI}} = 0$. The reference paper's figure stops below $I_{\mathrm{QCLI}} \approx 0.7$, so a high I_{QCLI} need not indicate exploitable structure and can simply indicate proximity to uniform.

E Additional arms

Data scaling. At $N \in \{10, 30, 50, 70, 90, 100\}$ the learned-direction IQP arm is flat for $N \geq 30$ (raw Conv2D-MMD^2 0.082–0.087) and so is the RBM (0.066–0.089), while the single-seed DCGAN leads already at N=10 (0.044 against 0.120) and swings by 3$\times$ non-monotonically (0.113 $\to$ 0.037), so the paper's pattern of a GAN catching up only as data grows is half reproduced.

Other arms. We ran 25 conditions, 16 IQP arms, the RBM and DCGAN baselines and seven zero-learning or PCA controls, of which the main text reports five IQP arms. The rest (Hamming-ordered latent block, other objectives and priors, a pseudo-time conditional arm) and the controls are released with the code under the same metrics. Every IQP arm keeps a nearest-neighbour distance to the held-out set (3.6–5.9) far above the split-half floor (0.95), so none is memorising the training set, and the M=8 arm collapses (PDF–JS 0.0103, MMD^2 0.244 at N=100).